STATISTICS IS THE GRAMMAR OF SCIENCE

PROBABILITY AND STATISTICS

LECTURE – 17

PROBABILITY DISTRIBUTIONS

EXPONENTIAL DISTRIBUTION

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EXPONENTIAL DISTRIBUTION

Exponential distribution is closely related to the Poisson distribution. Whereas the Poisson distribution is discrete and describes random occurrences over some interval, the **exponential distribution** is continuous and describes a probability distribution of the times between random occurrences. So *the exponential distribution is often used to measure the time that elapses between two occurrences of an event*.

EXAMPLES OF EXPONENTIAL DISTRIBUTION

Following are examples of random variables that might be exponential.

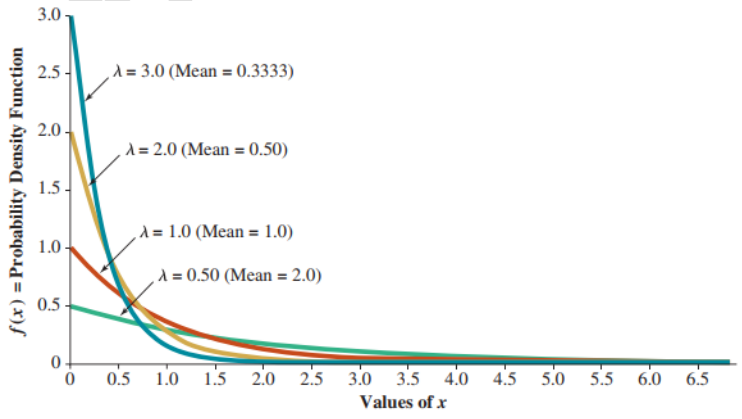
1. The duration of Jim's next telephone call
2. The time until the next crime in a certain town
3. The time until the next baby is born in a hospital
4. The time between two accidents at an intersection
5. The interarrival time between two customers at a post office
6. The time between two consecutive earthquakes in California

THE EXPONENTIAL DISTRIBUTION A continuous random variable X is said to have an exponential distribution if it has the following density function

$$f(x)=\begin{cases}\lambda e^{-\lambda x} & for\ x\geq o\\ 0 & otherwise\end{cases}$$

$where\ the\ \ parameter\ \lambda>0.$

GRAPHS OF SOME EXPONENTIAL DISTRIBUTIONS



PROPERTIES OF EXPONENTIAL DISTRIBUTION

1- The Mean of Exponential Distribution is $E(X)=1/\lambda$

2- The variance of Exponential Distribution is $V(X)=1/\lambda^2$

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EXAMPLES OF EXPONENTIAL DISTRIBUTION

IMPORTANT FORMULAS FOR CALCULATING PROBABILITIES

If X is an exponential random variable then its pdf is given by

$$f(x)=\begin{cases}\lambda e^{-\lambda x} & for\ x\geq o\\ 0 & otherwise\end{cases}$$

In this case following formulas are useful for calculating probabilities

$(a)\ P(a\leq X\leq b)=e^{-\lambda a}-e^{-\lambda b}$

$(b)\ P(X\geq a)=e^{-\lambda a}$

$(c)\ P(X\leq b)=1-e^{-\lambda b}$

EXAMPLE-1 Determine the following exponential probabilities.

$(a)\ P(X\leq 3|\lambda=0.68)$

$(b)\ P(X\geq 4|\lambda=1.7)$

$(c)\ P(3\leq X\leq 5|\lambda=1.35)$

SOLUTION Determine the following exponential probabilities.

$(a)\ P(X\leq 3|\lambda=0.68)$

$=1-e^{-\lambda b}$

$=1-e^{-(0.68)(3)}=0.8699$

$(b)\ P(X\geq 4|\lambda=1.7)$

$=e^{-\lambda a}=e^{-(1.7)(4)}=0.0011$

$(c)\ P(3\leq X\leq 5|\lambda=1.35)$

$=e^{-\lambda a}-e^{-\lambda b}$

$=e^{-(1.35)(3)}-e^{-(1.35)(5)}$

$=0.0162$

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EXAMPLE-2 A manufacturing firm has been involved in statistical quality control for several years. As part of the production process, parts are randomly selected and tested. From the records of these tests, it has been established that a defective part occurs in a pattern that is Poisson distributed on the average of 1.38 defects every 20 minutes during production runs. Use this information to determine the probability that less than 15 minutes will elapse between any two defects.

SOLUTION Let X be the time (in minutes) that will elapse between any two defects.

$Here\ \lambda = 2,\ Use\ \ 20\ \text{min}\ utes\ \text{int}\ erval.$

$we\ have\ to\ \ find\ \ P(X < 15/20) = P(X < 0.75) = ?$

$P(X < 0.75) = 1 - e^{-(1.38)(0.75)} = 0.6448$

EXAMPLE-3 At an intersection there are two accidents per day, on average. What is the probability that after the next accident there will be no accidents at all for the next two days?

SOLUTION Let X be the time (in days) between the next two accidents.

$Here\ \lambda = 2.\ we\ have\ to\ \ find\ \ P(X > 2) = ?$

$P(X > 2) = e^{-\lambda a} = e^{-(2)(2)} = e^{-4} = 0.02$

EXAMPLE-4 Suppose that every three months, on average, an earthquake occurs in California. What is the probability that the next earthquake occurs after three but before seven months?

SOLUTION Let X be the time (in months) until the next earthquake

$Here\ \lambda = 1/3.\ we\ have\ to\ \ find\ \ P(3 < X < 7) = ?$

$P(3 < X < 7) = e^{-\lambda a} - e^{-\lambda b} = e^{-(1/3)(3)} - e^{-(1/3)(7)} = 0.2709$

EXAMPLE-5 The Haines Services Company provides online chat support for small business customers who have purchased accounting software. It has been determined that the number of customers who attempt to connect to the chat service per hour is Poisson distributed with an average of 30 per hour. The time between connect requests is exponentially distributed with a mean time between requests of 2.0 minutes, At the current staffing level, if customer requests are too close together 45 seconds (0.75 minute) or less—the chat connection will fail and the customer will get a message saying to try again later. The managers at Haines are analyzing whether they should add additional service representatives to eliminate this problem. For this problem, find the probability that a customer will fail to connect.

SOLUTION Let X be the time (in minutes) a customer's wait to chat.

$Here\ \lambda = 30/60 = 0.50.\ we\ have\ to\ \ find\ \ P(X \le 0.75) = ?$

$P(X \le 0.75) = 1 - e^{-\lambda b} = 1 - e^{-(0.50)(0.75)} = 0.3127$

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IMPORTANT CONCEPTS AND FUNCTIONS IN PROBABILITY

DEFINITE INTEGRAL

$$\int_a^b f(x)dx = F(b) - F(a)$$

PROPERTIES OF DEFINITE INTEGRAL

1. $\int_a^a f(x)dx = 0$
2. $\int_a^b f(x)dx = -\int_b^a f(x)dx$
3. $\int_a^c f(x)dx = \int_a^b f(x)dx + \int_b^c f(x)dx$
4. $\int_{-a}^{a} f(x)dx = \begin{cases} 0, & \text{if } f(x) \text{ is an odd function} \\ 2\int_0^a f(x)dx, & \text{if } f(x) \text{ is an even function} \end{cases}$

MACLAURIN SERIES

$$f(x) = f(0) + \frac{x}{1!}f'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \ldots\ldots\ldots$$

MACLAURIN SERIES OF SOME IMPORTANT FUNCTIONS

1. $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \ldots\ldots$
2. $\frac{1}{1-x} = 1 + x + x^2 + x^3 + x^4 + x^5 + \ldots\ldots$

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GAMMA FUNCTION

$$\Gamma(n)=\int_0^{\infty} x^{n-1}e^{-x}dx \qquad \text{also} \qquad \int_0^{\infty} x^{n-1}e^{-ax}dx=\frac{\Gamma(n)}{a^n}$$

PROPERTIES OF GAMMA FUNCTION

1. $\Gamma(n)=(n-1).\Gamma(n-1)$
2. $\Gamma(n)=(n-1)!$ *If 'n' is a positive integer.*
3. $\Gamma(1)=1$
4. $\Gamma(\frac{1}{2})=\sqrt{\pi}$
5. $\Gamma(-\frac{1}{2})=\frac{1}{\sqrt{\pi}}$

EXAMPLES OF GAMMA FUNCTION

$$\int_0^{\infty} x\ e^{-x}\,dx=\Gamma(2)=1!=1$$

$$\int_0^{\infty} x^2\ e^{-x}\,dx=\Gamma(3)=2!=2$$

$$\int_0^{\infty} x^3\ e^{-2x}\,dx=\frac{\Gamma(4)}{2^4}=\frac{3!}{16}=\frac{6}{16}=\frac{3}{8}$$

$$\int_0^{\infty} x^5\ e^{-3x}\,dx=\frac{\Gamma(6)}{3^6}=\frac{5!}{729}=\frac{120}{729}$$

BETA FUNCTION

$$B(\alpha,\beta)=\int_0^1 x^{\alpha-1}(1-x)^{\beta-1}dx=\frac{\Gamma(\alpha).\Gamma(\beta)}{\Gamma(\alpha+\beta)}$$

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PROPERTIES OF EXPONENTIAL DISTRIBUTION

PROPERT-1 AREA UNDER THE EXPONENTIAL CURVE

We have to show that $\int_0^\infty f(x)dx = 1$

$$L.H.S = \int_0^\infty f(x)dx$$

$$= \int_0^\infty \lambda e^{-\lambda x}\, dx = \lambda \int_0^\infty e^{-\lambda x}\, dx = \lambda \left| \frac{e^{-\lambda x}}{-\lambda} \right|_0^\infty = (-1)[0-1] = 1 = R.H.S$$

PROPERT-2 MEAN OF EXPONENTIAL DISTRIBUTION

$$Mean = E(X) = \int_0^\infty x\, f(x)dx$$

$$= \int_0^\infty x \lambda e^{-\lambda x}\, dx = \lambda \int_0^\infty x e^{-\lambda x}\, dx = \lambda \frac{\Gamma(2)}{\lambda^2} = \frac{1!}{\lambda} = \frac{1}{\lambda}$$

PROPERT-3 VARIANCE OF EXPONENTIAL DISTRIBUTION

$$V(X) = E(X^2) - (E(X))^2 \text{ -------(1)}$$

we have $E(X) = \frac{1}{\lambda}$

Now $E(X^2) = \int_0^\infty x^2\, f(x)dx$

$$= \int_0^\infty x^2 \lambda e^{-\lambda x}\, dx = \lambda \int_0^\infty x^2 e^{-\lambda x}\, dx = \lambda \frac{\Gamma(3)}{\lambda^3} = \frac{2!}{\lambda^2} = \frac{2}{\lambda^2}$$

Put the values of $E(X)$ *and* $E(X^2)$ *in equation* (1), *we get*

$$V(X) = \frac{2}{\lambda^2} - \left(\frac{1}{\lambda}\right)^2$$

$$= \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}$$

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PROPERT-4 MOMENT GENERATING FUNCTION

Moment Generating Function is given by

$$M_X(t) = E\left(e^{tx}\right)$$

$$= \int_0^\infty e^{tx}\, f(x)dx$$

$$= \int_0^\infty e^{tx}\, \lambda e^{-\lambda x}\, dx = \lambda \int_0^\infty e^{tx}\, e^{-\lambda x}\, dx = \lambda \int_0^\infty e^{-(\lambda - t)x}\, dx$$

$$= \lambda \left| \frac{e^{-(\lambda - t)x}}{-(\lambda - t)} \right|_0^\infty = \frac{-\lambda}{\lambda - t}\left[0 - 1\right] = \frac{\lambda}{\lambda - t}$$

Mean using Moment Generating Function is given by

$$E(X) = \left[\frac{d}{dt} M_X(t)\right]_{t=0}$$

$$= \left[\frac{d}{dt}\left(\frac{\lambda}{\lambda - t}\right)\right]_{t=0}$$

$$= \left[\lambda(-1)(\lambda - t)^{-2}(-1)\right]_{t=0}$$

$$= \left[\lambda(\lambda - t)^{-2}\right]_{t=0} = \frac{1}{\lambda}$$

Variance using Moment Generating Function is given by

For variance we shall first find $E\left(X^2\right)$

$$E\left(X^2\right) = \left[\frac{d^2}{dt^2} M_X(t)\right]_{t=0}$$

$$= \left[\frac{d}{dt}\left(\lambda(\lambda - t)^{-2}\right)\right]_{t=0}$$

$$= \left[\lambda(-2)(\lambda - t)^{-2-1}(-1)\right]_{t=0} = \frac{2}{\lambda^2}$$

$$V(X) = E\left(X^2\right) - \left(E(X)\right)^2$$

$$= \frac{2}{\lambda^2} - \left(\frac{1}{\lambda}\right)^2 = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}$$

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PROPERT-5 CUMULANT GENERATING FUNCTION

Cumulant Generating Function is given by

$$K_X(t) = \ln(M_X(t))$$
$$= \ln\left(\frac{\lambda}{\lambda - t}\right)$$
$$= \ln\lambda - \ln(\lambda - t)$$

Mean using Cumulant Generating Function is given by

$$E(X) = \left[\frac{d}{dt}K_X(t)\right]_{t=0}$$
$$= \left[\frac{d}{dt}(\ln\lambda - \ln(\lambda - t))\right]_{t=0}$$
$$= \left[0 - \frac{1}{\lambda - t}\times -1\right]_{t=0}$$
$$= \left[\frac{1}{\lambda - t}\right]_{t=0}$$
$$= \frac{1}{\lambda}$$

Variance using Cumulant Generating Function is given by

$$V(X) = \left[\frac{d^2}{dt^2}M_X(t)\right]_{t=0}$$
$$= \left[\frac{d}{dt}\left(\frac{1}{\lambda - t}\right)\right]_{t=0}$$
$$= \left[(-1)(\lambda - t)^{-2}(-1)\right]_{t=0}$$
$$= \left[(\lambda - t)^{-2}\right]_{t=0}$$
$$= \frac{1}{\lambda^2}$$

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